
Continuous Memory Against Discrete Approximation:

The Out-of-Sample Forecasting Superiority of the Rough Fractional Stochastic Volatility Model Over Discrete Autoregressive Baselines

Semester Project Written Report

ECON 421 Econometrics

Author: Gorazd Atanasovski

Student ID: 101199791

Instructor: Dr. Mike Allgrunn

Submission Date: April 25, 2026

“Since we will never really know why the prices of financial assets move, we should at least make a faithful model of how they move.”

Jean-Philippe Bouchaud, Capital Fund Management

Abstract

In this semester project, I examine the out-of-sample forecasting power of the continuous Rough Fractional Stochastic Volatility (RFSV) model against two discrete industry-standard models, namely the AR model of order p , denoted $AR(p)$, and the Heterogeneous Autoregressive model, $HAR(3)$. My econometric hypothesis can be stated in the following way. Raw asset prices follow a non-stationary geometric random walk, therefore the direct use of regression analysis on price series is econometrically inefficient. The correct variable of interest is the underlying stochastic volatility process which displays a fractal geometry and is stationary. As instantaneous volatility is not observable, daily realized variance, calculated from the square sum of intra-day ultra-high-frequency log returns, serves as its proxy and is stabilized using logarithmic transformation to alleviate the problem of extreme right skewness and outliers introduced by market microstructure. Estimating the fractional scaling law based on sixteen years of S&P 500 ultra-high-frequency five-minute data results in $H \approx 0.14$, implying violent anti-persistence and roughness of the volatility surface.

The failure of autoregressive models in a rolling out-of-sample test stems from two sources: firstly, finite lag truncation destroys the memory property of the market across the chosen horizon; secondly, OLS estimation of the coefficients becomes very brittle under market stress, since the covariates become ill-conditioned and cause large swings in the regression coefficients. This problem is circumvented by the implementation of continuous fractional integration using the Riemann–Liouville integral, where the weight function does not change due to the universal constant H . In an out-of-sample backtest using a strict 500-day rolling window, RFSV obtains a P-Ratio of 0.561 at the 20-day horizon versus 0.610 for $AR(5)$ and 0.604 for $HAR(3)$.

Keywords: Rough Volatility · Fractional Brownian Motion · Hurst Exponent · Realized Variance · Rolling OLS · Coefficient Fragility · HAR Model · P-Ratio · Volterra Integral

Contents

1	Introduction and Economic Rationale	4
1.1	Econometric Target: Why Raw Prices Are Non-Stationary	4
1.2	Stationary Target: Latent Volatility	4
1.3	Isolation of the Signal: Construction of the Daily Realized Variance Measure	5
1.4	Stabilization of the Log-Variance Series	6
2	The Architecture of Time-Series Memory	7
2.1	Short Memory versus Long Memory: A Formal Definition	7
2.2	The Fractional Reality of Volatility Memory	8
3	Fractional Scaling Law and Formal Derivation	9
3.1	Fractional Brownian Motion: Formally Defined	9
3.2	Self-Similarity and the Fractional Scaling Law	9
3.3	Derivation of the Absolute Moment Equation	10
3.4	Logarithmic Linearization and OLS Estimation of H	11
3.5	Empirical Extraction and Physical Interpretation of $H \approx 0.14$	12
4	Econometric Baselines and Structural Inefficiencies	14
4.1	The Autoregressive $AR(p)$ Baseline	14
4.1.1	The Fatal Structural Problem: Hard Memory Truncation	14
4.2	The HAR(3): The Heterogeneous Autoregressive Baseline Model	15
4.2.1	The Fundamental Defect: Arithmetic Averaging Destroys Fractional Memory Structure	16
5	The RFSV Continuous Engine	17
5.1	The Volterra Integral Representation	17
5.2	Fractional Kernel: Continuous Power-Law Decay	18
6	Out-of-Sample Rolling Crucible and Coefficient Fragility	18
6.1	Rejection of In-Sample OLS Inference	18
6.2	Rolling Window Methodology	20
6.3	Coefficient Fragility: The Fundamental Econometric Failure Mechanism . .	20
6.4	Zero-Coefficient Estimation: The Structural Immunity of RFSV	21

7 Results and Conclusion	22
7.1 The P-Ratio as the Measure of Out-of-Sample Forecasting Accuracy	22
7.2 P-Ratio Performance Out-of-Sample: The Irrefutable Conclusion	23
7.3 Conclusion	25
References	26

1. Introduction and Economic Rationale

1.1 Econometric Target: Why Raw Prices Are Non-Stationary

The primary prerequisite for conducting rigorous multivariate regression within the context of quantitative finance is to identify a stationary target variable. Failure to do so will invalidate all future coefficients, standard errors, and forecasts.

By construction, each financial asset's log-price process $Y_t = \ln S_t$ follows a continuous-time semi-martingale process according to the physical measure $\mathbb{P}$, with its drift rate, Brownian motion component, and variance process being the economically most meaningful terms:

Log-Price Semi-Martingale Dynamics

$$dY_t = \mu_t dt + v_t dW_t \quad (1)$$

where μ_t is a drift term, W_t is a standard one-dimensional Brownian motion under $\mathbb{P}$, and v_t is the volatility process. The raw price level process $S_t = e^{Y_t}$ itself is a geometric random walk, whose unconditional mean and variance diverge as time progresses and whose properties are time-dependent. This process is said to be integrated of order one, denoted $I(1)$, requiring a single differentiation to obtain stationarity.

Running a multivariate regression on non-stationary variables will lead to spurious regression problems, whereby the covariate design matrix and dependent variable share a common stochastic trend, causing R^2 values to approach one while generating t -statistics which are significant without the existence of a meaningful forecasting relationship. Spurious regression has been well-established in the econometrics literature by Granger and Newbold (1974) and renders prices unsuitable for forecasting.

1.2 Stationary Target: Latent Volatility

As such, the task at hand is to rethink the regression target. Instead of focusing on trying to identify which direction the price will move in — which would not even be feasible due to the integration of the price process — one must instead analyze the latent volatility process v_t . Log-variance of price is a stationary process, mean-reverts at economically relevant horizons, and as shall be proven in the upcoming derivation displays a fascinating fractal structure exploitable for forecasting purposes via its measurement using high-frequency

observations and inclusion into a forecasting equation.

The economic rationale for the importance of the variance forecast lies in its relevance to financial decisions made under uncertainty: Black–Scholes–Merton derivatives pricing theory, Value-at-Risk calculations used by institutions in risk management, hedging ratio calculation, and volatility trading strategies are all heavily impacted by variance forecasting accuracy.

1.3 Isolation of the Signal: Construction of the Daily Realized Variance Measure

The first challenge in constructing a volatility forecast is that instantaneous volatility cannot be observed from the data. It is rather latent and requires econometric proxies to be constructed from it. Daily Realized Variance is the correct proxy for this task, constructed as a sum of intra-day ultra-high-frequency log returns squared.

The importance of logarithmic differencing should be underlined here. The five-minute log return over the i -th intra-day interval is defined as:

Intra-Day Log Return via Logarithmic Differencing

$$Y_{t_i} = \ln S_{t_i} - \ln S_{t_{i-1}} \quad (2)$$

Logarithmic differencing transforms the non-stationary price into the stationary measure — the proportional percentage movement of the asset. This transformation allows us to interpret a \$10 move as a shock of the same intensity whether the price of the underlying asset was \$100 or \$1,000 prior to it — i.e., 10% and 1%, correspondingly. The raw dollar measure of movement cannot tell us anything relevant about the intensity of the shock and thus cannot be used as an explanatory variable in our regression model. Instead, logarithmic differencing allows for normalizing price levels, assigning equal mathematical significance to shocks of identical relative intensity but at different levels of asset price.

In turn, Daily Realized Variance is created as a sum of all intra-day log returns' squared values:

Daily Realized Variance as a Proxy for Latent Volatility

$$\hat{v}_t = \sum_{i=1}^N Y_{t_i}^2 \quad (3)$$

The choice of squaring the return before the summation is not incidental — squaring plays a crucial role in the econometrics behind the measure. Summation of raw log returns over the period will inevitably produce a near-zero sum in the case of a volatile period characterized by both upward and downward price action. This means that violent moves are partially cancelled with their opposites when using simple summation. However, if we square each value, the result is an unbiased measure of total turbulence without regard to the sign of each event.

1.4 Stabilization of the Log-Variance Series

However, the untransformed variance series $\hat{v}_t$ cannot serve as the dependent variable in OLS regression. Two problems related to this decision arise.

Right-skewness and leptokurtosis. Markets display highly skewed behavior, particularly during crisis periods. The tail observations can become extremely large in magnitude as a response to a sovereign debt crisis, a sudden halt of market operations due to a pandemic, or changes in the policy of the central bank. Consequently, the variance series in these events becomes extremely large in magnitude compared to other observations, resulting in severe right-skewness with very high kurtosis.

OLS leverage and coefficient destabilization. In OLS multivariate regressions, extreme outliers in the dependent variable can exert undue influence on the objective function because it is sensitive to the magnitude of each observation's variance. An observation with a variance that is ten times as large as its mean, for example, contributes one hundred times more to the minimization criterion than does an average observation. For the OLS estimator to be able to minimize the criterion, it must shift its fitted coefficient vector to explain the outlier at the cost of misrepresenting other points. The result is an unstable and distorted coefficient vector that is inefficient and lacks generalizability to out-of-sample data.

To correct this distortion problem in econometric models, we should take the logarithm of the realized variance before using it in any form:

Logarithmic Stabilization of the Realized Variance Series

$$X_t = \ln(\hat{v}_t) \tag{4}$$

The logarithmic function is strictly concave, and therefore its slope declines with respect to an increase in input value. Consider two realizations of variance, where one is a

typical elevated day and the other is a crisis day: 10 and 1,000, respectively. In the raw data series, this ratio equals 100:1. However, after taking the logarithm of each number, the ratio reduces to about 4.6. In this way, the logarithm transforms a series with a violent right tail into a distribution whose left and right tails are balanced, i.e., approximately normally distributed. All further calculations are based on the log-variance series X_t in this project.

2. The Architecture of Time-Series Memory

2.1 Short Memory versus Long Memory: A Formal Definition

Under the classical definition of the memory property of a stationary time series, the concept depends on how fast the autocovariance function decreases when the lag grows indefinitely. Depending on its asymptotic behavior, the memory property influences how deeply the forecasting model must incorporate the information contained in the series.

Definition 1 (Short Memory). *A stationary time series $\{X_t\}$ has short memory if its autocovariance function declines exponentially quickly to zero as the lag Δ grows; in fact, so quickly that the sum of its values is absolutely convergent:*

Short Memory: Exponential Autocovariance Decay

$$\text{Cov}(X_t, X_{t+\Delta}) \propto e^{-\lambda\Delta}, \quad \lambda > 0 \quad (5)$$

It implies that the spectral density function is continuous and finite everywhere, including the zero-frequency point.

Definition 2 (Long Memory). *A stationary time series $\{X_t\}$ has long memory, also referred to as strong dependence, if its autocovariance function has a slow decrease that follows a power law and thus results in the divergence of the sum:*

Long Memory: Power-Law Autocovariance Decay

$$\text{Cov}(X_t, X_{t+\Delta}) \propto \Delta^{-\gamma}, \quad 0 < \gamma < 1 \quad (6)$$

Its spectral density has a pole at the origin of frequency.

This distinction is critical when considering the specification of forecasting models.

With a short-memory process, any econometric information is contained only within a small number of lags, and anything further back in time is not considered to be meaningful information for the forecast. In a long-memory process, on the other hand, there is enough predictive correlation between past and future to consider all of the process' history relevant to some extent for the forecast. Conventional autoregressive models assume a sharp cut-off for lags and are thus designed to represent processes with short memory and cannot possibly account for long memory by construction.

2.2 The Fractional Reality of Volatility Memory

In practice, volatility itself falls somewhere in between the two types of processes. Its logarithmic process follows the fractional Ornstein–Uhlenbeck process, with an autocovariance function that decays at a fractional exponential rate. According to Gatheral et al. (2018), the true autocovariance function of the RFSV process has the following specific form:

RFSV Autocovariance: Fractional Exponential Decay Structure

$$\text{Cov}(X_t, X_{t+\Delta}) \approx C e^{-B\Delta^{2H}} - D \quad (7)$$

where B , C , and D are positive scalar constants $\in \Gamma$ determined by the model parameters and the Hurst exponent H .

If $H = 0.5$, then the decay rate corresponds to an ordinary Ornstein–Uhlenbeck process and, hence, to short memory. If $H < 0.5$, which is empirically supported, the autocovariance decay rate is much slower than exponential, with the key difference being that the autocovariance function decays at a slower-than-exponential rate. This means that in terms of persistence, the autocovariance function displays much more weight and persistence than that for the usual short-memory process.

Such slower-than-exponential decay replicates the behavior of classical long memory at a finite, yet empirically measurable, horizon. In particular, it results in persistent correlations at both weekly and monthly scales, making it behave superficially similar to a power-law process associated with long memory. However, since the autocovariance function is, in fact, integrable, the process cannot formally be defined as long memory. Thus, the RFSV volatility model represents a structurally different class of models, falling somewhere between short memory and long memory processes, with conventional estimation procedures often classifying it incorrectly. This erroneous classification has actually been used to prove long memory in volatility modeling in the long-memory forecasting literature. In particular, applying usual tests to the RFSV volatility process generates spurious long

memory, as confirmed by simulation in Gatheral et al.

3. Fractional Scaling Law and Formal Derivation

3.1 Fractional Brownian Motion: Formally Defined

The underlying element of the RFSV approach is the concept of Fractional Brownian Motion (fBm). Let $\Gamma \subset \mathbb{R}$ denote the scalar field over which the process is indexed. A *fractional Brownian motion* $\{W_t^H\}_{t \in \Gamma}$ with Hurst parameter $H \in (0, 1)$ is defined as a centered Gaussian process with the following covariance structure:

Covariance Structure of Fractional Brownian Motion

$$\mathbb{E}[W_t^H W_s^H] = \frac{1}{2} \left(|t|^{2H} + |s|^{2H} - |t - s|^{2H} \right), \quad t, s \in \Gamma \quad (8)$$

In particular, when $H = 0.5$, the covariance function of fBm is similar to the one of the ordinary Brownian motion process with independent increments. For any other $H \neq 0.5$, increments of the fBm process become correlated. Positive correlation corresponds to persistence ($H > 0.5$), while anti-persistence occurs if $H < 0.5$. The latter case is applicable in volatility modeling.

3.2 Self-Similarity and the Fractional Scaling Law

The essential structural characteristic of the fBm process is *self-similarity*. In particular, a process of fBm maintains its statistical distribution when subjected to a special transformation of time and scale (amplitude). For any scalar $c \in \Gamma$ with $c > 0$:

Self-Similarity Property of Fractional Brownian Motion

$$\{W_{ct}^H\}_{t \geq 0} \stackrel{d}{=} \{c^H W_t^H\}_{t \geq 0} \quad (9)$$

where $\stackrel{d}{=}$ denotes equality in distribution. Self-similarity implies an important connection between the distribution of any fBm increment in a certain lag and the same increment over the unit lag. This relationship becomes the very basis of the RFSV empirical calibration procedure:

Fractional Scaling Law for fBm Increments

$$W_{t+\Delta}^H - W_t^H \stackrel{d}{=} \Delta^H W_1^H \quad (10)$$

It means that for any lag value Δ , an fBm increment can be expressed in terms of the increment over the unit lag, scaled with respect to the lag and raised to the power of the Hurst exponent H .

3.3 Derivation of the Absolute Moment Equation

Under the RFSV setup, the log-variance process X_t is modulated by fBm, so that $X_t \approx 2\nu W_t^H + C$ for scalar constants $\nu \in \Gamma$ (the volatility-of-volatility) and $C \in \Gamma$ (a centering constant). In this setting, the absolute moment of the log-variance difference for a particular lag can be calculated analytically.

Proposition 1. *Let X_t be driven by fBm W_t^H with Hurst exponent H and volatility-of-volatility $\nu \in \Gamma$. Then for any moment order $q > 0$:*

Absolute Moment Scaling of Log-Variance Increments

$$\mathbb{E}[|X_{t+\Delta} - X_t|^q] = K_q \nu^q \Delta^{qH} \quad (11)$$

Proof. **Step 1.** Invoke the self-similarity of fBm. By equation (8):

Step 1: Distributional Equivalence via Self-Similarity

$$X_{t+\Delta} - X_t = 2\nu(W_{t+\Delta}^H - W_t^H) \stackrel{d}{=} 2\nu \cdot \Delta^H \cdot W_1^H \quad (12)$$

Step 2. Since W_1^H is a standard Gaussian random variable, define $Z = W_1^H \sim \mathcal{N}(0, 1)$. Then:

Step 2: Reduction to Standard Gaussian Absolute Value

$$|X_{t+\Delta} - X_t| \stackrel{d}{=} 2\nu \Delta^H |Z| \quad (13)$$

Step 3. Apply the expectation operator:

Step 3: Expectation Operator Applied to the Scaled Gaussian

$$\mathbb{E}[|X_{t+\Delta} - X_t|^q] = \mathbb{E}[(2\nu \Delta^H |Z|)^q] = (2\nu)^q \Delta^{qH} \mathbb{E}[|Z|^q] \quad (14)$$

Step 4. Define the scalar constant $K_q \in \Gamma$ as the q -th absolute moment of the standard Gaussian distribution, expressed via the Euler Gamma function $\Gamma(\cdot)$ evaluated at the relevant scalar argument:

Gaussian Absolute Moment via the Euler Gamma Function

$$K_q = \mathbb{E}[|Z|^q] = \frac{2^{q/2} \Gamma\left(\frac{q+1}{2}\right)}{\sqrt{\pi}} \quad (15)$$

This is a pure mathematical constant with no economic content, determined solely by the Gaussian distributional assumption and the moment order q .

Step 5. Absorbing $(2)^q$ into the baseline energy scalar $\nu^q \in \Gamma$ and collecting terms yields the claimed result:

Final Result: Power-Law Moment Equation for Log-Variance Increments

$$\mathbb{E}[|X_{t+\Delta} - X_t|^q] = K_q \nu^q \Delta^{qH} \quad (16)$$

□

In turn, each component of this moment equation has a clear economic interpretation without any redundancy. First, the standard Gaussian moment $K_q \in \Gamma$ is an unambiguous mathematical quantity defined exclusively based on the assumption of normality and the selected moment order. This part bears no economic significance whatsoever. Then, the energy scalar $\nu^q \in \Gamma$ characterizes the magnitude of the volatility-of-volatility behavior, namely the market-specific measure that governs the scale of fractional variance increments. Finally, the scaling factor Δ^{qH} describes the fractal structure of log-variance increments in relation to the lengthening time lag Δ . The whole econometric challenge boils down to determining the value of H using this equation.

3.4 Logarithmic Linearization and OLS Estimation of H

Estimating H directly from this non-linear relationship is computationally impractical and numerically unstable. The usual procedure consists of taking logarithms on both sides of the equation, which converts this equation into a linear regression.

First, define the empirical q -th moment of the log-variance increments at lag Δ :

Empirical Absolute Moment of Log-Variance Increments

$$m(q, \Delta) = \frac{1}{N - \Delta} \sum_{t=1}^{N-\Delta} |X_{t+\Delta} - X_t|^q \quad (17)$$

Taking logarithms of equation (12) and substituting the empirical moment for the expectation, with $\beta_0 = \ln(K_q \nu^q)$ absorbing all constants, we obtain a bivariate linear regression problem in log-log space:

Log-Log Linearization for OLS Estimation of the Hurst Exponent

$$\ln m(q, \Delta) = \beta_0 + (qH) \cdot \ln \Delta \quad (18)$$

Specifically, the dependent variable corresponds to the logarithm of the empirical moment, the explanatory one to $\ln \Delta$, the intercept accounts for all constants, and the slope coefficient is equal to the product of the moment order q and H . Using Ordinary Least Squares, we derive the estimator of the slope and divide it by the moment order, thereby obtaining the value of H :

OLS Estimator of the Hurst Exponent from the Log-Log Slope

$$\hat{H} = \frac{\hat{\beta}_1}{q} \quad (19)$$

It is vital to note the distinction between the epistemological purpose of OLS in this case and its common econometric function. Typically, the purpose of regression analysis is to check if there is any statistical relationship between two variables, which is evaluated through p -values and t -tests. RFSV calibration does not use this approach at all. The law of fractional scaling is a geometric property of the volatility surface revealed by log-log linearity in data. Hence, OLS works as an exact measuring tool that mechanically measures the slope of a graph already present in empirical data.

3.5 Empirical Extraction and Physical Interpretation of $H \approx 0.14$

The empirical estimate of H around 0.14, applied to the same sample of realized variances from sixteen years of five-minute S&P 500 daily data (4,636 observations), shows that the resulting Hurst parameter estimates vary negligibly with respect to the choice of moment order, being stable at all moments examined. Table 1 presents the full calibration.

Table 1: Hurst Exponent Calibration by Moment Order — S&P 500, 2006–2022

Moment Order (q)	OLS Slope ($\hat{\beta}_1 = q\hat{H}$)	Extracted $\hat{H} = \hat{\beta}_1/q$
0.5	0.0683	0.1365
1.0	0.1350	0.1350
1.5	0.2013	0.1342
2.0	0.2676	0.1338
Average	—	$H \approx 0.1349 \approx 0.14$

The range of H across moment orders is under 0.003, small enough for any econometric purpose. Above all, however, the perfect coincidence of all regression lines drawn in the log-log plot is decisive proof of the monofractality hypothesis: the roughness structure of the volatility surface of the S&P 500 exhibits statistical self-similarity at all scales and levels of magnification of the time series under study. If there had been either multifractality or no self-similarity, then H would have changed with moment order; but it does not.

Physical interpretation of $H \approx 0.14$. The importance of H cannot be fully appreciated without understanding the implications of H taking on other values.

For standard Brownian motion ($H = 0.5$), successive increments are independent; the increment of the process within any interval reveals no information about future increments of the process. This property is known in the finance literature as memorylessness, representing the strong version of the Efficient Market Hypothesis (EMH).

When $H > 0.5$, successive increments are positively correlated, implying a tendency for persistent trends; the assumption behind early fractional volatility models (such as those used by Comte and Renault) that $H > 0.5$ was precisely this one.

When $H < 0.5$ — the regime confirmed for the S&P 500 — successive increments are negatively correlated; above-average increments tend to be followed by below-average or negative increments. The process is anti-persistent: it reverses direction repeatedly, regardless of scale. Its sample path is jagged and rough, rather than smooth. In increments there is no directional dependence; there is no predictability at the micro level, although there is mean reversion and macro-level fractal persistence.

$H = 0.14$ is not just a value lower than 0.5; it is a value far lower than 0.5. This degree

of anti-persistence is very strong, as witnessed by the Hölder continuity of its sample path: log-volatility is nowhere differentiable and is, moreover, rougher than the path of a standard Brownian process at every scale. As such, it represents the key empirical discovery of the RFSV literature: the fact that the world’s most liquid equity index is rougher than Brownian motion is an important stylized fact.

4. Econometric Baselines and Structural Inefficiencies

4.1 The Autoregressive AR(p) Baseline

In classic time-series econometrics, the Autoregressive approach is considered the basis upon which all future models can be built. The AR(p) specification for the log-variance forecasting problem takes the form:

AR(p) Specification for Log-Variance Forecasting

$$\log \sigma_{t+\Delta}^2 = K_0^\Delta + \sum_{i=0}^p C_i^\Delta \log \sigma_{t-i}^2 + \varepsilon_{t+\Delta} \quad (20)$$

where K_0^Δ is a horizon-specific intercept and C_i^Δ is the coefficient weight assigned to the i -th lag of the log-variance. Its coefficients are estimated via minimization of the in-sample sum of squared residuals using the OLS estimator:

Ordinary Least Squares Coefficient Estimator

$$\hat{\mathbf{C}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (21)$$

where $\mathbf{X}$ is the $T \times (p+2)$ covariate design matrix containing a column of ones and the lagged log-variance covariates, and $\mathbf{y}$ is the vector of forward log-variance observations.

4.1.1 The Fatal Structural Problem: Hard Memory Truncation

The key econometric problem with the AR specification — which will hereinafter be termed *hard memory truncation* — is due to setting a predetermined lag. After the AR coefficients are found, the model is forced to assign zero probability to any observations beyond the lag p ; it does not asymptotically approach a small coefficient but assigns an explicit zero:

Hard Memory Truncation: Explicit Zero Weight Beyond Lag p

$$C_i^\Delta = 0 \quad \forall i > p \quad (22)$$

It should be understood that in doing so, $\text{AR}(p)$ specifies an exact statement that the history of log-variance beyond a certain number of trading days carries zero informational content regarding tomorrow's variance. Technically speaking, this specification means that the market has finite memory.

The above specification contradicts empirical reality. Estimating the Hurst exponent based on sixteen years of five-minute market data suggests a fractional scaling law, meaning the memory of log-volatility extends across multiple weeks and months. This process indicates that a shock to the volatility surface affects it in a fractionally integrated way, persisting at a rate specified by H . The hard memory wall of the AR model abruptly stops the propagation of effects.

On the practical side, it leads to the so-called phase lag — a certain gap between the true market and the model in its ability to react to changes in the volatility surface. Indeed, when there is an unexpected event — a credit event, a geopolitical conflict, or some other change — a couple of trading days will be required for the model to accumulate enough samples in its lookback window before adjusting its forecast. Practically, during this adjustment period, the model's prediction lags behind the true market development. Essentially, it tells its user about yesterday's market situation while the true market is already developing a tomorrow's state. Trading with such a forecast means incurring real financial losses due to misforecast risk.

4.2 The HAR(3): The Heterogeneous Autoregressive Baseline Model

The Heterogeneous Autoregressive model attempts to incorporate long-term memory dynamics using a very concise specification consisting of three components only. The economic interpretation is rooted in the heterogeneous market hypothesis, according to which there exist multiple types of market actors — individual investors making intraday trades, institutional investment funds operating on a daily level, and pension funds with multi-year horizons — that contribute to market development simultaneously. The three components represent volatility memory dynamics for one day, one week, and one month, respectively:

HAR(3) Heterogeneous Autoregressive Specification

$$\begin{aligned}
\log \sigma_{t+\Delta}^2 &= K_0^\Delta + C_0^\Delta \log \sigma_t^2 \\
&\quad + C_5^\Delta \cdot \frac{1}{5} \sum_{i=0}^4 \log \sigma_{t-i}^2 \\
&\quad + C_{20}^\Delta \cdot \frac{1}{20} \sum_{i=0}^{19} \log \sigma_{t-i}^2 + \varepsilon_{t+\Delta}
\end{aligned} \tag{23}$$

The weekly component averages the log-variance for the last five trading days, while the monthly component averages the log-variance for the last twenty trading days.

4.2.1 The Fundamental Defect: Arithmetic Averaging Destroys Fractional Memory Structure

While the HAR model represents an important theoretical breakthrough in contrast to the AR model due to the extended memory of twenty trading days, the model contains an important econometric flaw in its treatment of this memory structure: the weekly and monthly components are *unweighted arithmetic averages*.

Stated succinctly, the monthly component takes all observations equally, meaning that each observation is treated as having equal weight:

HAR Monthly Component: Equal-Weight Arithmetic Averaging

$$\frac{1}{20} \sum_{i=0}^{19} \log \sigma_{t-i}^2 = \frac{1}{20} [\log \sigma_t^2 + \log \sigma_{t-1}^2 + \cdots + \log \sigma_{t-19}^2] \tag{24}$$

For instance, the log-variance observed today is treated mathematically with the same weight as the log-variance observed nineteen days ago. However, the fundamental scaling law of fractional calculus dictates that the weight decreases continuously as a power of the lag, specifically $\propto u^{-(H+0.5)}$. Therefore, the HAR model contradicts this law since each observation is given equal weight, which leads to the loss of high-frequency memory structure.

The problem can be summarized succinctly: in estimating the memory structure of volatility using a twenty-day block, the HAR model averages together the information-rich recent observations with less-recent observations into a single entity. As a result, the HAR model loses the ability to detect regime shifts in volatility within the block. A sudden

change in volatility will be smoothed out to a mild increase in the block and treated as noise over the full twenty-day window.

Moreover, the cut-offs involved — five trading days for the weekly model and twenty trading days for the monthly one — are completely arbitrary calendar rules defined by humans. The fractional power-law scaling does not care or know about people having five-day weeks and twenty-day months. Volatility memory fades continuously and in fractional increments; there are no stop-go events occurring after five days followed by a fresh start after twenty days.

5. The RFSV Continuous Engine

5.1 The Volterra Integral Representation

In order to address the two deficiencies of the AR and HAR approaches in terms of memory and averaging, the RFSV model makes a single architectural change: it switches from a discrete summation to a continuous integration approach based on the conditional expectation of fBm increments under the Riemann–Liouville fractional integral representation. As such, the RFSV forecasting equation becomes that of a Volterra-type affine approximation:

RFSV Forecasting Equation: Riemann–Liouville Volterra Integral

$$\mathbb{E}[\log \sigma_{t+\Delta}^2 \mid \mathcal{F}_t] \approx \frac{\cos(H\pi)}{\pi} \int_0^r \frac{\log v_{t-\Delta u}^2}{(u+1)u^{H+0.5}} du \quad (25)$$

where $\mathcal{F}_t$ denotes the natural filtration generated by the log-volatility process up to time t , and r is a truncation parameter chosen such that the tail integral beyond r is negligible for a specified precision tolerance ε .

Such integral equations contain the unknown function entering into the equation linearly. In our case, this corresponds to the fractional weighting of the history of observed variance through the kernel of the integral, which arises as the result of approximating the fractional Ornstein–Uhlenbeck process governing log-volatility with fBm locally when the mean-reverting coefficient is small relative to the forecasting horizon. As it was empirically established, this is indeed the case for all standard forecasting horizons.

5.2 Fractional Kernel: Continuous Power-Law Decay

From the standpoint of forecasting, the main practical feature of the RFSV model is associated with its integral representation and is embedded into the denominator in particular. The fractional kernel is defined as:

Fractional Memory Kernel: Continuous Power-Law Decay Function

$$\mathcal{K}(u; H) = \frac{1}{(u + 1) u^{H+0.5}} \quad (26)$$

This fractional kernel requires further analysis. First and foremost, the function is monotonically decreasing, meaning that the most recently observed log-variance receives the largest weight, whereas more distant realizations become gradually less and less influential.

Compared to the HAR model, the main difference here is the continuity of the decay function and its full dependence on the value of H . Contrary to HAR, the RFSV model involves no artificial cut-off points or blocks where all weights are set equally and no arbitrarily defined equal-weights window. Instead, for each moment of the past continuum of log-variance values, a corresponding weight determined by H is used without exceptions and truncations.

Also, the kernel value never becomes precisely zero at any finite distance. Unlike the AR model, which sets its tails to precisely zero after a specific lag cut-off point, the RFSV integral maintains a link to the entire observable history of the log-variance that is non-zero, but continuously decaying. The fractal memory effects of market shocks far away in the past continue to play their fractional role in shaping the forecasts, as opposed to being simply chopped off at the cut-off point.

6. Out-of-Sample Rolling Crucible and Coefficient Fragility

6.1 Rejection of In-Sample OLS Inference

Any analysis of a model based on in-sample criteria like R^2 , F -statistics, and p -values calculated over the entire period of historical data cannot be used as any form of validation of the predictive ability of a forecasting model.

Fitting an OLS regression to the entire historical sample means that the regression

estimator uses the entire time frame of data from start to finish, including periods which are in the future relative to any given forecasting date within the sample. The weight placed by the OLS estimator on the log-variance on any particular day is based on information from future periods. The 2009 recovery, for instance, influences the OLS estimator in terms of assigning appropriate weights to the events of the 2008 financial crisis. This is what is referred to as lookahead bias — the incorporation of information from the future into explanations of past occurrences. The result of such an estimator is a reflection of mere memorization and not forward-looking predictive accuracy.

Using classical econometrics, the performance of the baselines is excellent in-sample. All coefficients are estimated with impressive statistical significance. Table 2 documents these results in full. However, they are only indicative of the memorization ability of the models and give no insight into how the models would perform prospectively.

Table 2: In-Sample OLS Regression Results — AR(5) and HAR(3) Models

Panel A: AR(5) $R^2 = 0.723$ $F = 2,423$ $p(F) = 0.000$ $N = 4,636$				
Variable	Coeff.	Std. Err.	t -stat	p -value ¹
Intercept (K_0)	−0.778	0.084	−9.24	0.000
Lag 1 (C_1)	0.442	0.015	30.15	0.000
Lag 2 (C_2)	0.208	0.016	13.03	0.000
Lag 3 (C_3)	0.095	0.016	5.85	0.000
Lag 4 (C_4)	0.097	0.016	6.05	0.000
Lag 5 (C_5)	0.080	0.015	5.46	0.000
Panel B: HAR(3) $R^2 = 0.726$ $F = 4,072$ $p(F) = 0.000$ $N = 4,621$				
Variable	Coeff.	Std. Err.	t -stat	p -value ¹
Intercept (K_0)	−0.485	0.090	−5.38	0.000
Daily (C_0)	0.358	0.018	20.42	0.000
Weekly (C_5)	0.398	0.028	14.24	0.000
Monthly (C_{20})	0.195	0.023	8.60	0.000

¹ denotes significance at the 1% level.

6.2 Rolling Window Methodology

Proper out-of-sample performance testing requires that the forecasting model face the same information constraints that it would face when put into practical use. Specifically, at any forecasting date, the forecasting model is permitted to utilize only observations that come before that forecasting date. Such constraints are imposed via a strictly trailing 500-day rolling window.

The method works as follows. In every instance of a forecast, only the last 500 trading days will comprise the estimation window. A new regression model will be estimated based on that window of data. Based on this new model, we construct the covariate design matrix $\mathbf{X}$, perform an OLS regression and extract the coefficient vector $\hat{\mathbf{C}}$, which is then used to make the forecast at the target horizon Δ . The forecast error is tracked. The analysis window slides by one day, discards the oldest observation, adds a new one at the end, and repeats the whole process. This step-by-step procedure goes through thousands of iterations in the out-of-sample evaluation period.

Using a 500-day analysis window is a careful consideration from an econometric perspective. It is long enough for us to estimate a statistically solid sample given normal market conditions — leading to a numerically stable inversion of the OLS covariate design matrix $(\mathbf{X}^T \mathbf{X})^{-1}$. Meanwhile, the window remains short enough not to induce a fallacy of stability due to the use of such a long window that spans different volatility regimes of the market history.

6.3 Coefficient Fragility: The Fundamental Econometric Failure Mechanism

A 500-day window setting brings to light the fundamental econometric problem with real-world application of the discrete autoregressive models: the problem of fragile coefficients.

During the period of relatively calm markets, the OLS estimation procedure goes off without any problems. There is homogeneity in the distribution of observations over time; the realizations of the volatility proxy are mostly close to their historical mean values, and the covariate design matrix $\mathbf{X}$ is well-conditioned and nearly stationary, allowing for numerically stable matrix inversion and smooth behavior of the resulting coefficient vector from one estimation cycle to another.

A significant shift toward stressed market conditions dramatically worsens the situation and turns it catastrophic. A system-wide event causing increased volatility will lead to a large number of extreme observations over the rolling window. They play the role of high-leverage outliers in the context of our estimation procedure — they lie far away from

the center of mass of the covariate design matrix $\mathbf{X}$, and they exert disproportionate influence over the OLS criterion compared to normal observations.

As a result, the covariance matrix condition deteriorates. With every incoming extreme observation, the cross-product matrix $\mathbf{X}^T \mathbf{X}$ becomes increasingly ill-conditioned: its condition number $\kappa(\mathbf{X}^T \mathbf{X}) = \lambda_{\max}/\lambda_{\min}$ grows large as the smallest eigenvalue $\lambda_{\min}$ approaches zero. Matrix inversion of such an ill-conditioned matrix is prone to arbitrarily large fluctuations of the output even with minor perturbations of the input, as a relative perturbation δ in the data can produce a relative change of up to $\kappa \cdot \delta$ in the inverted matrix $(\mathbf{X}^T \mathbf{X})^{-1}$ and hence in $\hat{\mathbf{C}}$.

The consequence is dramatic instability in the coefficients of the vector from one rolling forecast to the next. The coefficient of any one of the lags will shift dramatically as a proportion of its value over just two consecutive days, with the introduction and subsequent exit of a crisis observation into the rolling window. The twenty-day block of the HAR model exhibits extreme volatility around its minimum and maximum values as the period of the crisis enters and exits the twenty-day average.

These oscillating and unstable coefficients produce incoherent forecasts, which have no relation whatsoever to the underlying dynamics of the variance process. They generate structural forecasting error in the form of extremely inaccurate forecasts at exactly the point where accuracy is of greatest economic importance.

This is not due to poor choices in the parameters or too short a window. This is an inevitable outcome of using OLS. Stable estimates can only be achieved with well-conditioned covariates in the design matrix, which does not occur under conditions of high-volatility regimes. Fine tuning and adjusting the lag structure is not enough to protect discrete autoregressive models against the failure mode since it occurs due to the inversion of the OLS design matrix $(\mathbf{X}^T \mathbf{X})^{-1}$.

6.4 Zero-Coefficient Estimation: The Structural Immunity of RFSV

Because the forecasting equation of the RFSV model contains no free parameters which need to be estimated at each forecasting step, the model is structurally immune to coefficient fragility. The Hurst exponent H , being a physical constant of the volatility process, is obtained once at calibration of the model across the entire history of data using the deterministic OLS solution described in Section 3. There are no lag coefficients C_i^Δ to estimate, no covariate design matrix $\mathbf{X}$ to construct, and no matrix inversion $(\mathbf{X}^T \mathbf{X})^{-1}$ to perform at any rolling step.

In every single forecasting step, the RFSV model simply applies its fixed fractional kernel $\mathcal{K}(u; H)$ to the rolling window of observed log-variance, computing the Riemann sum approximation of the integral in equation (15). All weights for the kernel are determined by the value of $H = 0.14$ and are the same at all forecasting steps regardless of whether the market state is one of stability or system-wide financial crisis.

The advantage of the RFSV model in terms of econometrics lies in the following idea: the primary reason for the high forecast error in case of discrete autoregressive processes is not the potential model misspecification itself, but the instability of the rolling OLS method due to the market environment. Thus, avoiding this step and directly fixing the structure of weights with respect to the constant H makes it possible to eliminate the largest cause of forecast error compared to alternative specifications.

Universality of H as the Physical Constant is important. Numerous empirical analyses have demonstrated that the value of the Hurst exponent lies within a certain, quite narrow range in the case of most global stock indices, fixed-income volatility surfaces, commodities, and foreign exchange rate measures of volatilities ($H \in [0.06, 0.20]$). The constancy of this value is maintained throughout time sub-periods and volatility metrics, regardless of moment orders. Such a near-universal constancy of the value of H gives sufficient grounds to consider it a stable property of the market that can be used in model specification.

7. Results and Conclusion

7.1 The P-Ratio as the Measure of Out-of-Sample Forecasting Accuracy

In order to avoid in-sample biases when assessing the ability of the model to make accurate forecasts on new data, the criterion of evaluation should be introduced. The P-Ratio is defined as the ratio of the model forecasting error to the global variance of the series of observations. The numerator of the P-Ratio is the sum of squared forecasting errors, whereas its denominator is the total sum of squares of deviations of the target from its average:

P-Ratio: Out-of-Sample Forecasting Accuracy Criterion

$$P = \frac{\sum_k (X_{k+\Delta} - \hat{X}_{k+\Delta})^2}{\sum_k (X_{k+\Delta} - \bar{X})^2} \quad (27)$$

where $X_{k+\Delta}$ is the realized log-variance at horizon Δ from evaluation step k , $\hat{X}_{k+\Delta}$ is the model's point forecast, and $\bar{X}$ is the empirical mean of the log-variance over the full evaluation period.

Thus, the P-Ratio equals the mean squared error divided by the variance of the observed series. The value of the P-Ratio of zero corresponds to the ideal theoretical forecast. The P-Ratio of unity means that the model forecast is equivalent to a naive forecast that suggests taking into account only the mean value of the historical series. Values between zero and unity indicate various levels of the success of prediction; smaller values of P-Ratio correspond to more accurate forecasts.

The P-Ratio is considered the analog of in-sample R^2 calculated for out-of-sample forecasts without any contamination with lookahead bias and over-fitting. Thus, it is an unbiased assessment of model prediction skills.

7.2 P-Ratio Performance Out-of-Sample: The Irrefutable Conclusion

Table 3 presents the complete out-of-sample P-Ratio matrix evaluated under the strict 500-day trailing rolling window for all three model specifications and three forecasting horizons.

Table 3: Out-of-Sample P-Ratio Matrix — 500-Day Rolling Window Evaluation

Horizon (Δ)	AR(5)	AR(10)	HAR(3)	RFSV
$\Delta = 1$ Day	0.273	0.273	0.269	0.267
$\Delta = 5$ Days	0.427	0.419	0.415	0.403
$\Delta = 20$ Days	0.610	0.611	0.604	0.561
Improvement over HAR(3) at $\Delta = 20$: -0.043				

Bold entries denote the dominant model at each horizon. Lower P-Ratio = superior forecasting accuracy.

One-Day Horizon. All models cluster within a narrow P-Ratio range for the one-day forecasting horizon. In particular, the RFSV model yields the minimum P-Ratio, slightly better than the discrete models. Increasing the lag depth does not yield any additional out-of-sample signal. The tight clustering is reflective of the economic truth: all models adequately capture the strong day-to-day autocorrelation of the log-variance series that serves as the primary short-horizon signal. The advantage of the RFSV model in this regard comes from its continuous integral formulation being particularly sensitive to fine-scale intraday dynamics.

Five-Day Horizon. The difference between RFSV and the baseline discrete models is noticeably greater for the five-day horizon. The explanation for this observation is the beginning of rolling coefficient instability in the OLS specifications. The estimation window needs to simultaneously capture persistent volatility regimes spanning several weeks while continuously incorporating the day-to-day changes to the covariate vector. As such, the coefficients for the RFSV model remain unchanged through the process, preserving the stability and predictive ability of the model.

Twenty-Day Horizon. The conclusive empirical demonstration is delivered by the twenty-day forecast horizon. Here, the deterioration of the AR models is dramatic. No additional lag terms can improve their performance at such a long horizon due to the nature of the signal decay in the AR process. Even the HAR model which, by design, accounts for multi-period memory and is the industry benchmark in predicting realized variances produces inferior results.

Here, we see that at a twenty-day horizon, the discrete models fail to explain almost half of the natural log-variance variation. Moreover, the repeated recalculation of the OLS regression coefficients at this long horizon creates a very ill-conditioned covariate matrix. Therefore, the estimated rolling parameters start drifting away from the actual values as daily updates introduce more noise into the calculations.

By contrast, the RFSV model provides considerable improvements over the baselines. This gain is not negligible, as at the institutional scale it amounts to structural improvements in risk-adjusted performance metrics. It also demonstrates a clear positive relationship with the forecast horizon length and consistently improves without resorting to complex, computationally expensive, and unreliable rolling coefficient re-estimation.

7.3 Conclusion

In this semester project, a comprehensive theoretical econometric model is proposed for variance forecasting based on continuous fractional integration against discrete autoregressive frameworks. Four independent factors are discussed in a sequential manner, forming a solid econometric argument, which requires no further justification.

The necessity of switching from raw prices' stationarity to log-variance, estimated using the Daily Realized Variance $\hat{v}_t = \sum_{i=1}^N Y_{t_i}^2$ and stabilized via the log-transformation $X_t = \ln(\hat{v}_t)$, stems directly from the price-level's non-stationarity, as well as the invisibility of spot volatility and the inherent skewness of the raw variance distribution.

Secondly, it is proven that the calibrated scaling law for the log-volatility process of S&P 500 corresponds to a Hurst exponent substantially below the Brownian one ($H \approx 0.14 \ll 0.5$), being robust for moments of any order. Furthermore, in line with the universal result for rough volatility across all asset classes, the log-volatility surface of S&P 500 exhibits anti-persistence behavior and an extremely rough fractal structure.

Discrete autoregressive specifications face two major pathological challenges: firstly, the finite memory horizon in AR and HAR models leads to a significant phase lag, due to the truncation of the market fractional memory at a constant time period. Secondly, the ill-conditioning of the covariates design matrix causes the coefficients to undergo extreme fluctuations, as captured formally through the OLS estimator $\hat{\mathbf{C}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ becoming numerically unstable.

Finally, the RFSV model mitigates these shortcomings simultaneously: first, the continuous framework removes the need for the coefficient vector estimation and, secondly, eliminates the issue with OLS matrix conditioning by replacing summation with the Riemann–Liouville fractional integral with the kernel depending only on H universally:

$$\mathbb{E}[\log \sigma_{t+\Delta}^2 \mid \mathcal{F}_t] \approx \frac{\cos(H\pi)}{\pi} \int_0^r \frac{\log v_{t-\Delta u}^2}{(u+1) u^{H+0.5}} du$$

In conclusion, it should come as no surprise that RFSV dominates over both short and long forecasting intervals ($P = 0.267$ at $\Delta = 1$ day versus $P = 0.561$ at $\Delta = 20$ days). The larger the time interval, the greater the advantage, as expected from the theoretical framework of this paper. With a short forecasting interval, a discrete approximation may be able to approximate the daily autocorrelation signal in the volatility series adequately. However, when dealing with the fractal memory of the volatility surface over longer time horizons, only a continuous approximation based on fractional integration can capture its

geometry.

It would be a gross misstatement to say that multivariate regression is a fundamentally flawed analysis methodology in general. In fact, in the case of stationary short-memory processes, an autoregressive approach remains both econometrically justified and computationally efficient. On the other hand, this paper has argued that the task of forecasting the fractally rough, anti-persistent, and long-range dependent log-variance of financial instruments over multiple days requires an econometric treatment beyond the capabilities of discrete OLS-based methodologies. In other words, whereas the fractional integration model using the empirical Hurst exponent can be regarded as an improvement on the traditional autoregressive model, the former constitutes the correct econometric approach to a fundamentally different type of stochastic process.

References

- [1] Andersen, T.G., Bollerslev, T., Diebold, F.X., and Labys, P. (2001). The Distribution of Realized Exchange Rate Volatility. *Journal of the American Statistical Association*, 96(453), 42–55.
- [2] Comte, F. and Renault, E. (1998). Long Memory in Continuous-Time Stochastic Volatility Models. *Mathematical Finance*, 8(4), 291–323.
- [3] Corsi, F. (2009). A Simple Approximate Long-Memory Model of Realized Volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- [4] Ding, Z., Granger, C.W.J., and Engle, R.F. (1993). A Long Memory Property of Stock Market Returns and a New Model. *Journal of Empirical Finance*, 1(1), 83–106.
- [5] Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is Rough. *Quantitative Finance*, 18(6), 933–949.
- [6] Granger, C.W.J. and Newbold, P. (1974). Spurious Regressions in Econometrics. *Journal of Econometrics*, 2(2), 111–120.
- [7] Mandelbrot, B.B. (1963). The Variation of Certain Speculative Prices. *The Journal of Business*, 36(4), 394–419.
- [8] Mandelbrot, B.B. and Van Ness, J.W. (1968). Fractional Brownian Motions, Fractional

Noises and Applications. *SIAM Review*, 10(4), 422–437.

- [9] Pipiras, V. and Taqqu, M.S. (2002). Deconvolution of Fractional Brownian Motion. *Journal of Time Series Analysis*, 23(4), 487–501.